COM105 Engineering Physics (I)

Tutorial 2

1. A student at Xiamen University Malaysia went to Xiamen, China first and then come back to the campus in Malaysia. A one-way trip is about 1900 km. How about his/her traveling distance and displacement?

2. Define the average/instantaneous velocity and acceleration.

3. In an experiment, a shearwater (a seabird) was taken from its nest, flown 5150 km away, and released. The bird found its way back to its nest 13.5 days after release. If we place the origin at the nest and extend the +x-axis to the release point, what was the bird's average velocity in m/s (a) for the return flight and (b) for the whole episode, from leaving the nest to returning?

4. Starting from a pillar, you run 200 m east (the +x-direction) at an average speed of 5.0 m/s and then run 280 m west at an average speed of 4.0 m/s to a post. Calculate (a) your average speed from pillar to post and (b) your average velocity from pillar to post.

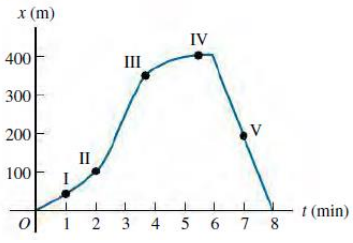
5. A Honda Civic travels in a straight line along a road. The car's distance x from a stop sign is given as a function of time t by the equation $x(t) = at^2 - bt^3$, where a = 1.50 m/s^2 and b = 0.0500 m/s^3. Calculate the average velocity of the car for each time interval: (a) t = 0 to t = 2.00 s; (b) t = 0 to t = 4.00 s; (c) t = 2.00 s to t = 4.00 s.

6. A car is stopped at a traffic light. It then travels along a straight road such that its distance from the light is given by $x(t) = bt^2 - ct^3$, where b = 2.40 m/s^2 and c = 0.120 m/s^3. (a) Calculate the average velocity of the car for the time interval t = 0 to t = 10.0 s. (b) Calculate the instantaneous velocity of the car at t = 0, t = 5.0 s, and t = 10.0 s. (c) How long after starting from rest is the car again at rest?

7. A bird is flying due east. Its distance from a tall building is given by $x(t)$ = 28.0 m + (12.4 m/s)t – (0.0450 m/s^3)t^3. What is the instantaneous velocity of the bird when t = 8.00 s?

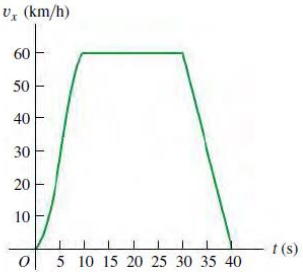
8. A physics professor leaves her house and walks along the sidewalk toward campus. After 5 min it starts to rain, and she returns home. Her distance from her house as a function of time is shown in Fig. E2.10. At which of the labeled points is her velocity (a) zero? (b) constant and positive? (c) constant and negative? (d) increasing in magnitude? (e) decreasing in magnitude?

Figure **E2.10**



9. Figure E2.12 shows the velocity of a solar-powered car as a function of time. The driver accelerates from a stop sign, cruises for 20 s at a constant speed of 60 km/h, and then brakes to come to a stop 40 s after leaving the stop sign. (a) Compute the average acceleration during these time intervals: (i) t = 0 to t = 10 s; (ii) t = 30 s to t = 40 s; (iii) t = 10 s to t = 30 s; (iv) t = 0 to t = 40 s. (b) What is the instantaneous acceleration at t = 20 s and at t = 35 s?

Figure **E2.12**



10. The position of the front bumper of a test car under microprocessor control is given by $x(t)$ = 2.17 m + (4.80 m/s^2) t^2 – (0.100 m/s^6) t^6. (a) Find its position and acceleration at the instants when the car has zero velocity. (b) Draw x-t, $v_{x\text{-}t}$, and $a_{x\text{-}t}$ graphs for the motion of the bumper between t = 0 and t = 2.00 s.

11. The fastest measured pitched baseball left the pitcher's hand at a speed of 45.0 m/s. If the pitcher was in contact with the ball over a distance of 1.50 m and produced constant acceleration, (a) what acceleration did he give the ball, and (b) how much time did it take him to pitch it?

12. The human body can survive an acceleration trauma incident (sudden stop) if the magnitude of the acceleration is less than 250 m/s^2. If you are in an automobile accident with an initial speed of 105 km/h and are stopped by an airbag that inflates from the dashboard, over what distance must the airbag stop you for you to survive the crash?

13. A juggler throws a bowling pin straight up with an initial speed of 8.20 m/s. How much time elapses until the bowling pin returns to the juggler's hand?

14. The acceleration of a bus is given by $a_x(t) = \alpha t$, where α = 1.2 m/s^3. (a) If the bus's velocity at time t = 1.0 s is 5.0 m/s, what is its velocity at time t = 2.0 s? (b) If the bus's position at time t = 1.0 s is 6.0 m, what is its position at time t = 2.0 s? (c) Sketch a_y-t, v_y-t, and x-t graphs for the motion.